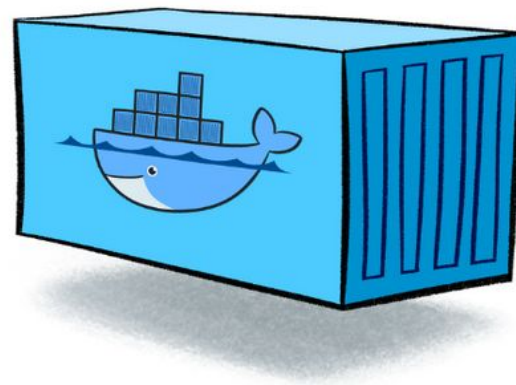


Containerization

QLSC 612 - May 2026
Sebastian Urchs



Teaching material adapted from Alyssa Dai adapted from Sebastian Urchs

Containers?



What we will cover

1. **Why** are containers useful for researchers?
2. Using and building containers with **Docker**
3. Using containers on supercomputers with **Singularity**

Why isolate environments?

Document your software environment



1: to share with colleagues



Document your software environment



1: to share with colleagues



The screenshot shows the GitHub repository for 'cog-align' by neurodatascience. The repository description states: 'This repository contains code for running the alignment for cognition project. It builds on code from the recent project. An empirical evaluation of functional alignment using inter-subject decoding. Thomas Bauriedl, Elizabeth DuPon, Jean-Baptiste Poline, & Bernard Thirion. doi: 10.1101/2020.12.07.415000'. The 'Requirements' section lists dependencies: fmralign, nibabel>=3.1, numpy>=1.18, matplotlib, pandas, scipy, and scikit-learn. The 'Installation' section provides instructions: 'First, make sure you have installed all the dependencies listed above. Then you can install cog-align by running the following commands:'. The 'Contributors' section lists: emdage (Elizabeth DuPon), thomasbauriedl (Thomas Bauriedl), jbpoline (Jean-Baptiste Poline), and jpoline (Jean-Baptiste Poline). The 'Languages' section shows Python (100.0%).

Requirements

Dependencies :

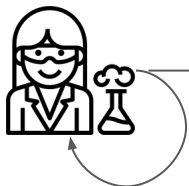
- `fmralign`
- `nibabel>=3.1`
- `numpy>=1.18`
- `matplotlib`
- `pandas`
- `scipy`
- `scikit-learn`

Installation

First, make sure you have installed all the dependencies listed above. Then you can install cog-align by running the following commands:

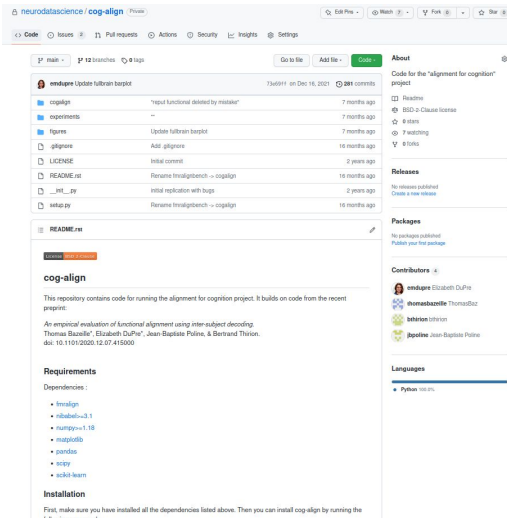
```
git clone https://github.com/neurodatascience/cog-align
cd cog-align
pip install -e .
```

Document your software environment



1: to share with colleagues

2: for yourself 3 months (days?) from now!



Requirements

Dependencies :

- **fmalign**
- **nibabel>=3.1**
- **numpy>=1.18**
- **matplotlib**
- **pandas**
- **scipy**
- **scikit-learn**

Installation

First, make sure you have installed all the dependencies listed above. Then you can install cog-align by running the following commands:

```
git clone https://github.com/neurodatascience/cog-align
cd cog-align
pip install -e .
```

Don't use the same environment for all projects



EVERY CHANGE BREAKS SOMEONE'S WORKFLOW.

Changing dependencies may do unexpected things

- A. Your updated the dependencies of an existing project

Don't use the same environment for all projects

Changing dependencies may do unexpected things

```
mriqc / pyproject.toml
Code Blame 235 lines (207 loc)
18 ]
19 dependencies = [
20     "acres",
21     "dipy >= 1.10.0",
22     'importlib_resources; p
23     "markupsafe ~ 2.0.1",
24     "matplotlib",
25     "migas >= 0.4.0",
26     "mriqc-learn",
27     "nibabel >= 3.0.1",
28     "nilearn >= 0.5.1",
29     "nipy >= 1.4",
30     "nireports ~ 24.0.2",
31     "nitransforms ~ 24.0",
32     "niworkflows ~1.10.1",
33     "numpy >= 1.20"

fmripred / pyproject.toml
Code Blame 206 lines (185 loc)
22 dependencies = [
23     "acres >= 0.2.0",
24     "looseversion >= 1.3",
25     "nibabel >= 4.0.1",
26     "nipy >= 1.8.5",
27     "nireports >= 24.1.0",
28     "nitime >= 0.9",
29     "nitransforms >= 24.1.1",
30     "niworkflows >= 1.12.2",
31     "numpy >= 1.24",
32     "packaging >= 24",
33     "pandas >= 1.2",
34     "psutil >= 5.4",
35     "pybids >= 0.16",
```

- A. Your updated the dependencies of an existing project
- B. Two projects use the same environment but need different versions of some dependencies

Reminder: Do not install things into your system Python!

Common installation issues

Installing into the system Python on Linux

On Linux systems, a Python installation will typically be included as part of the distribution. Installing into this Python installation requires root access to the system, **and may interfere with the operation of the system package manager and other components** of the system if a component is unexpectedly upgraded using `pip`.

On such systems, it is often better to use a virtual environment or a per-user installation when installing packages with `pip`.

```
[surchs@marvin ~]$ which python
/usr/bin/python
[surchs@marvin ~]$ which pip
/usr/bin/pip
[surchs@marvin ~]$
```

Consider: a cake

Home · Cakes · Perfect Cream Cheese Pound Cake

Perfect Cream Cheese Pound Cake

Published by [Sally](#) on February 18, 2019 - [700 comments](#)



not sponsored, but I absolutely adore Nordic ware bundt pans.
10-12 cups of batter. [This one](#) is also gorgeous! 😊

- 5 **Bake:** Bake the cream cheese pound cake at 325°F (163°C). Hal the cake with aluminum foil to prevent over-browning.
- 6 **Cool, then invert:** Let the pound cool for about 2 hours in the plate and cool completely before serving.

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Home · Cakes · Perfect Cream Cheese Pound Cake

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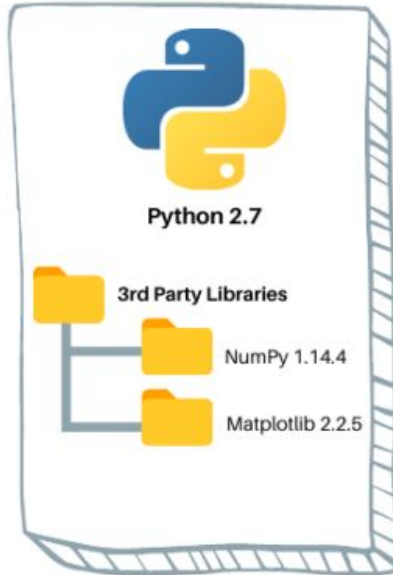


Isolate environments to handle different requirements

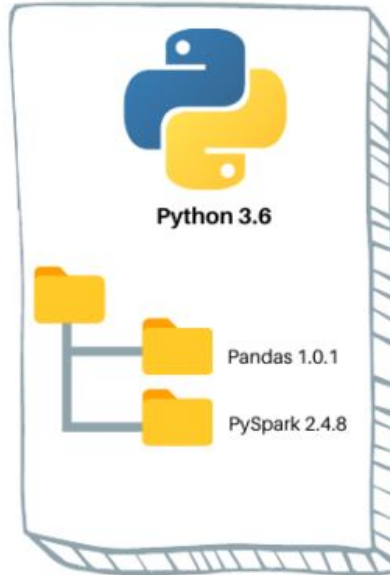


Isolate Python environments

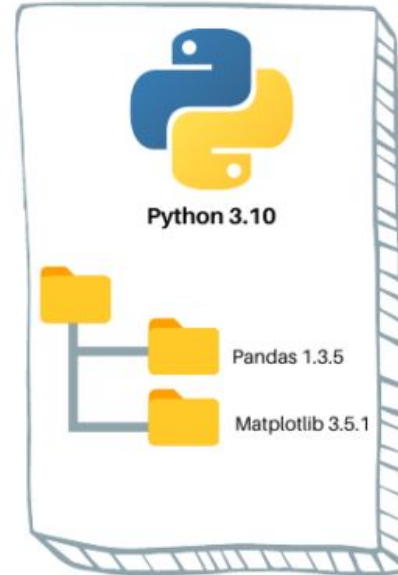
Virtual Environment 1



Virtual Environment 2



Virtual Environment 3



Why not just Python virtual environments?

External Dependencies

fMRIPrep is written using Python 3.8 (or above), and is based on [nipy](#).

fMRIPrep requires some other neuroimaging software tools that are not handled by the Python's packaging system (Pypi) used to deploy the `fmripred` package:

- [FSL](#) (version 6.0.5.1)
- [ANTs](#) (version 2.3.3 - NeuroDocker build)
- [AFNI](#) (version 22.3.06)
- [C3D](#) (version 1.3.0)
- [FreeSurfer](#) (version 7.3.2)
- [ICA-AROMA](#) (version 0.4.5)
- [bids-validator](#) (version 1.8.0)
- [connectome-workbench](#) (version 1.5.0)

- Not all dependencies are supported in Anaconda
- Not everything is written in Python or R
- Your Operating System (**OS**) also has packages and a package manager
- The same version problems apply to these

Need to capture the (entire) compute environment

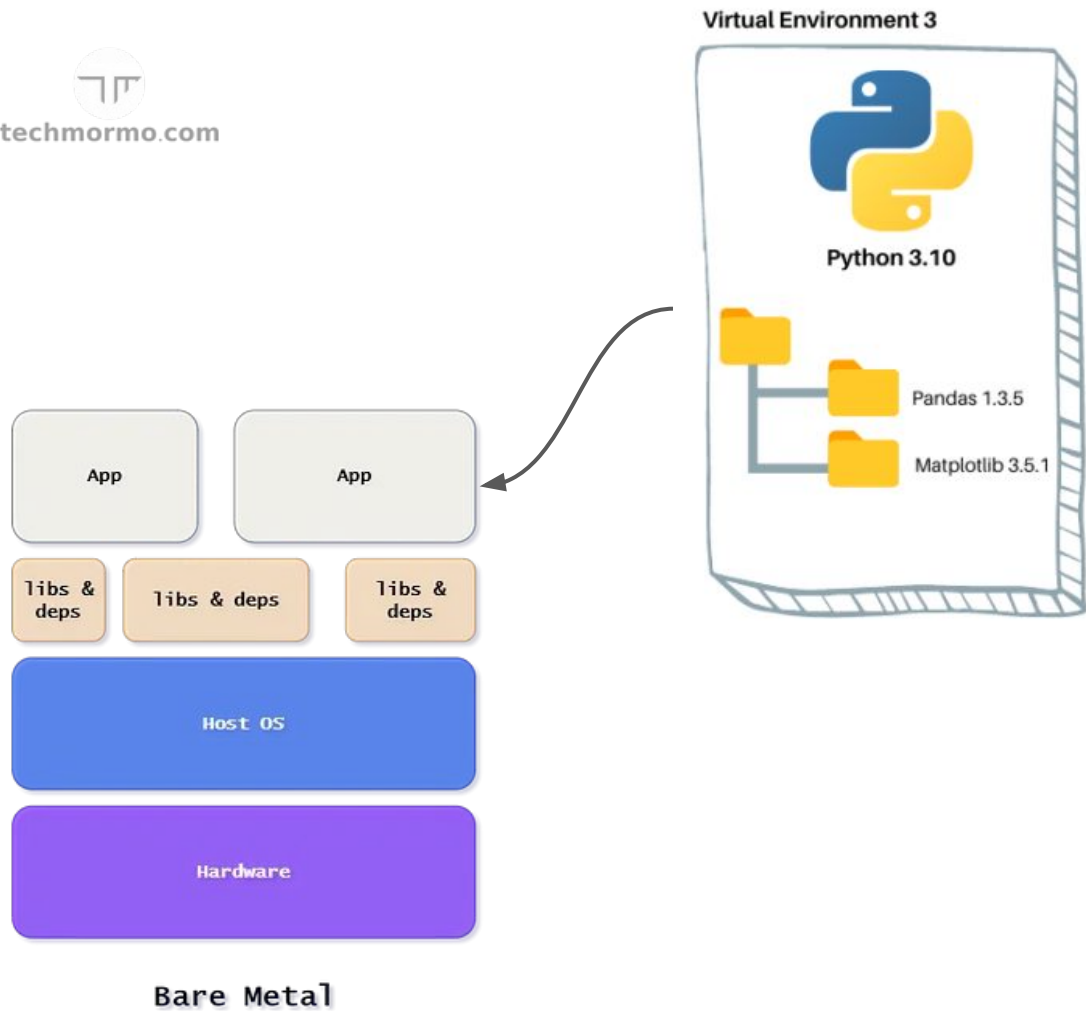




techmormo.com



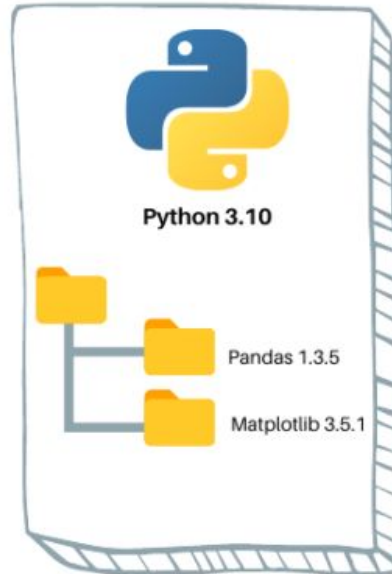
Bare Metal





Bare Metal

Virtual Environment 3

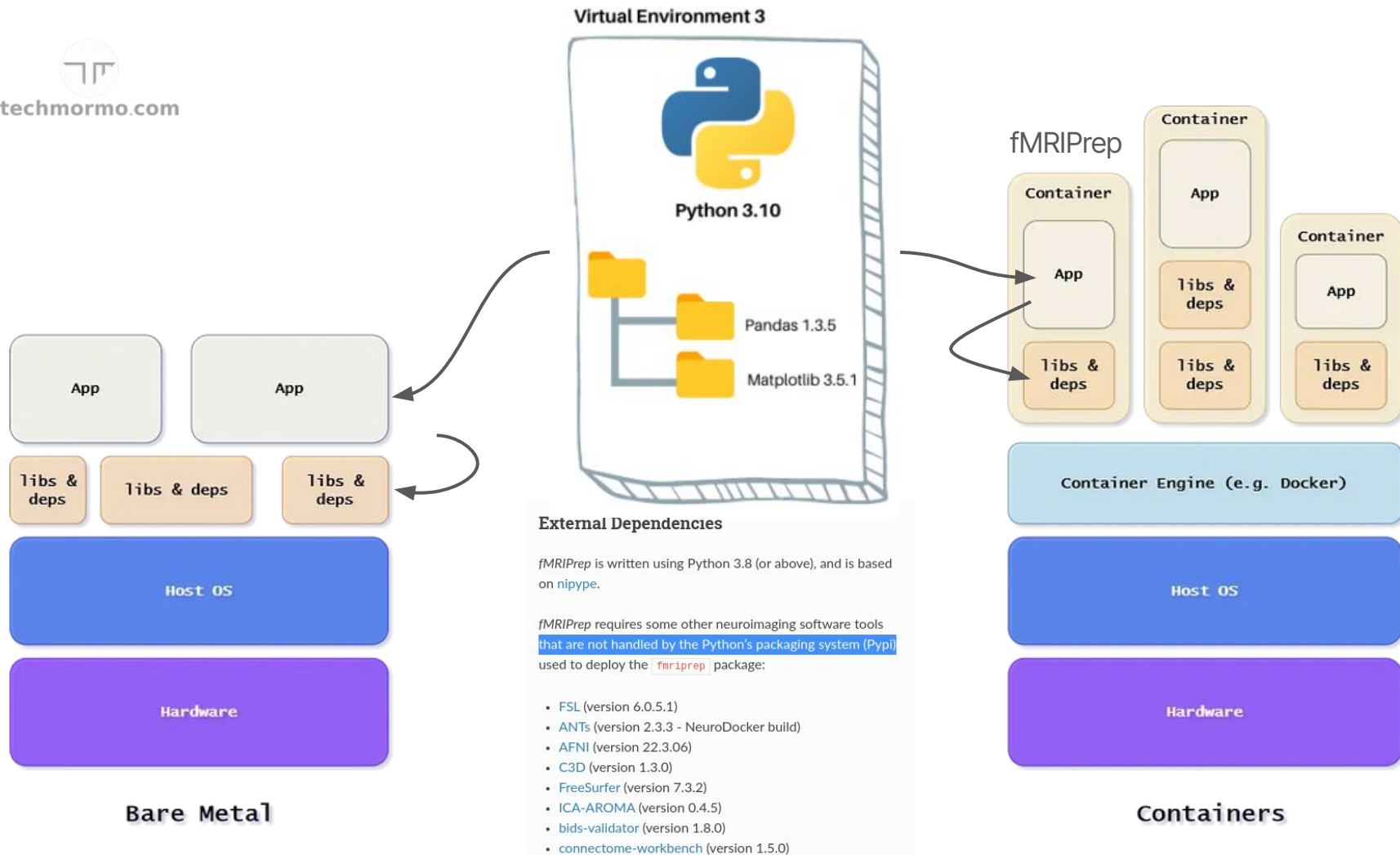


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- [connectome-workbench](#) (version 1.5.0)



Containers

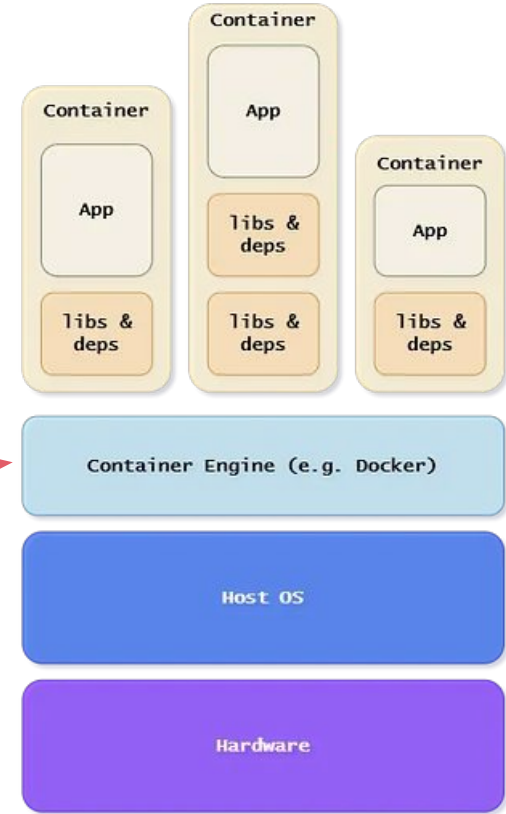
Containers

What are they

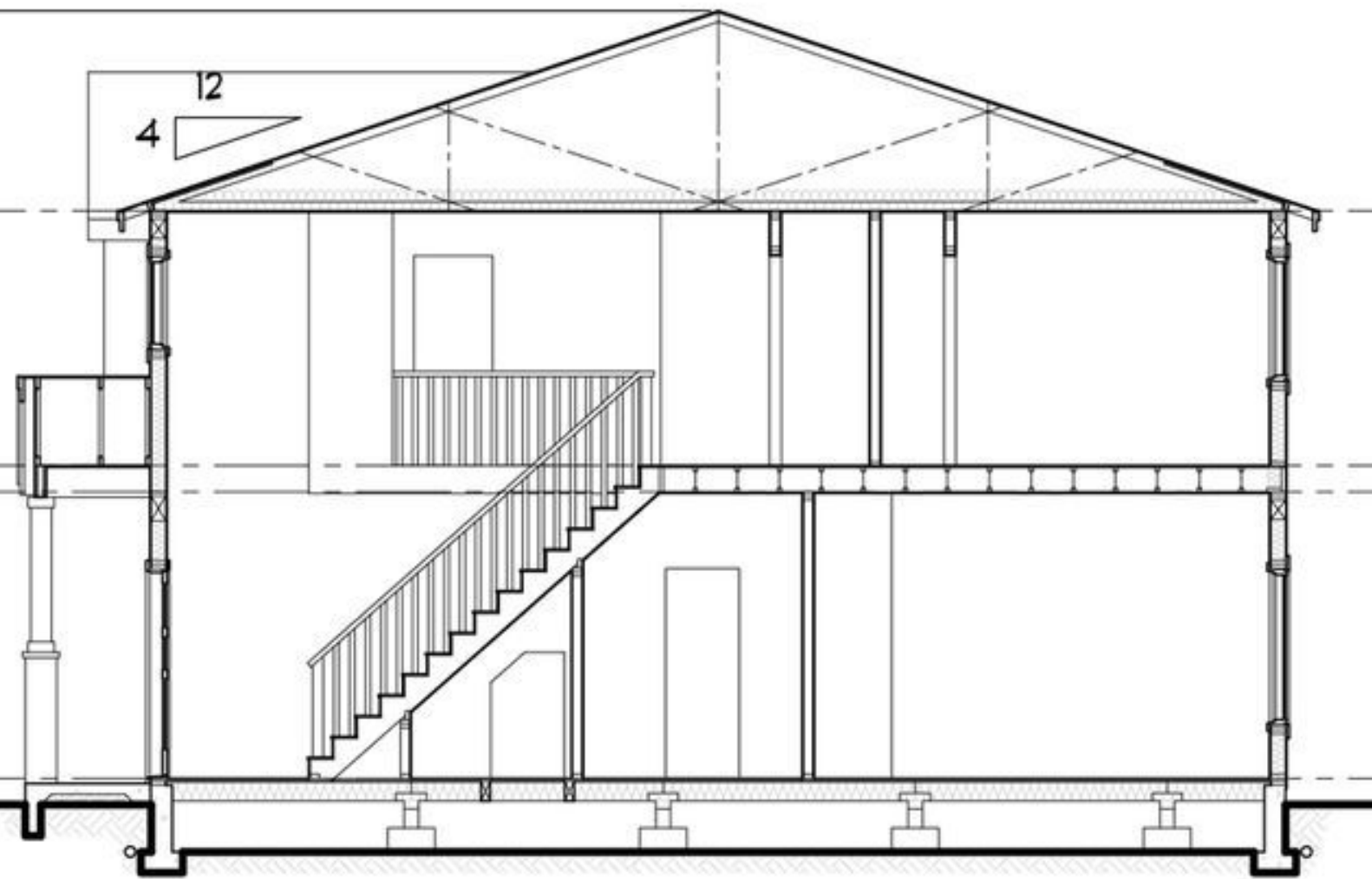
- isolated environments sharing the same host kernel / OS → **(OS virtualization)**
- from the inside, a container looks like a separate computer, can't see outside
- within each container, you can have your desired binary and library dependencies

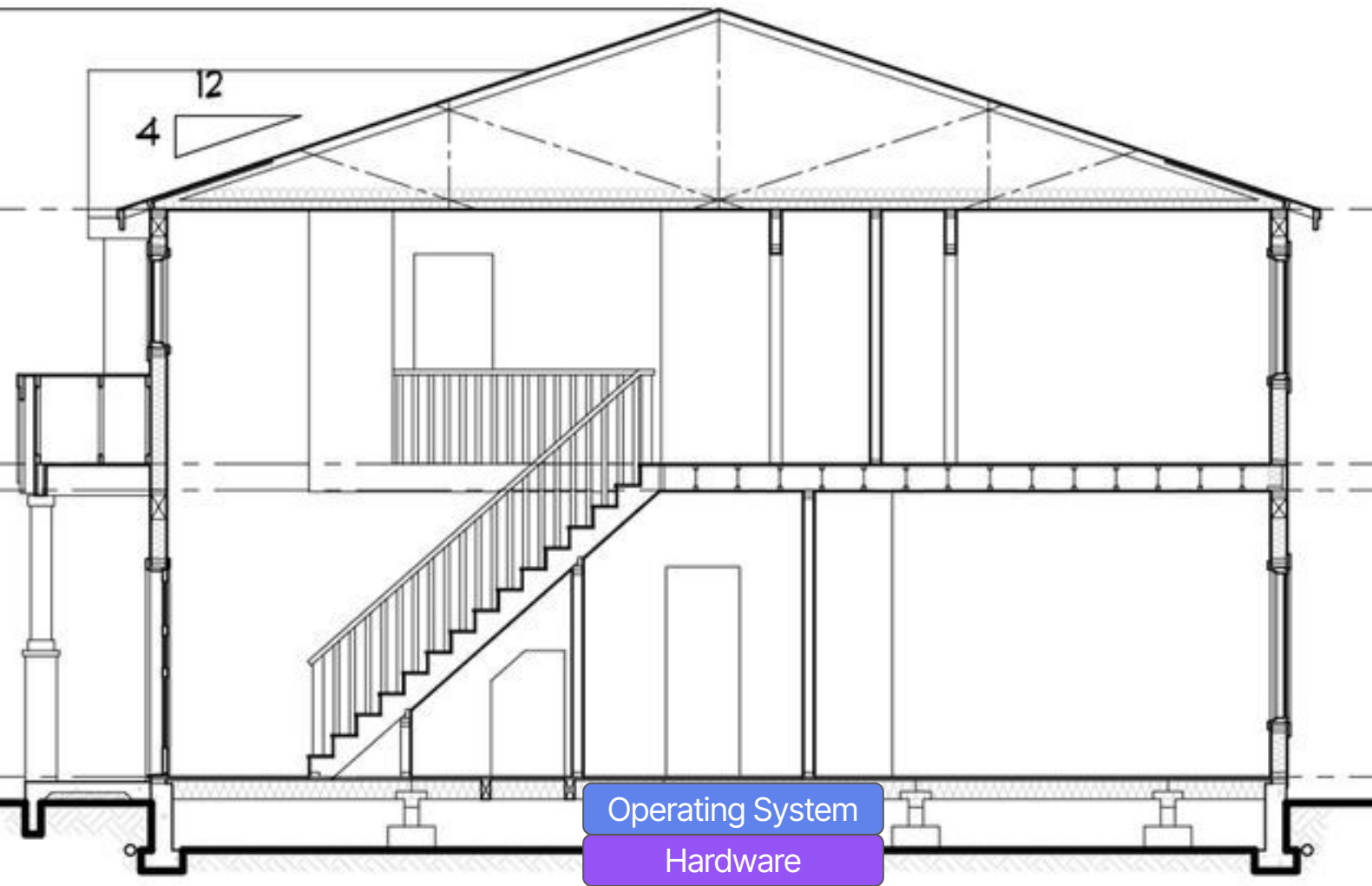
How do I make one

- use a container engine
- **Docker** is the most widely used
- Singularity is used on supercomputers



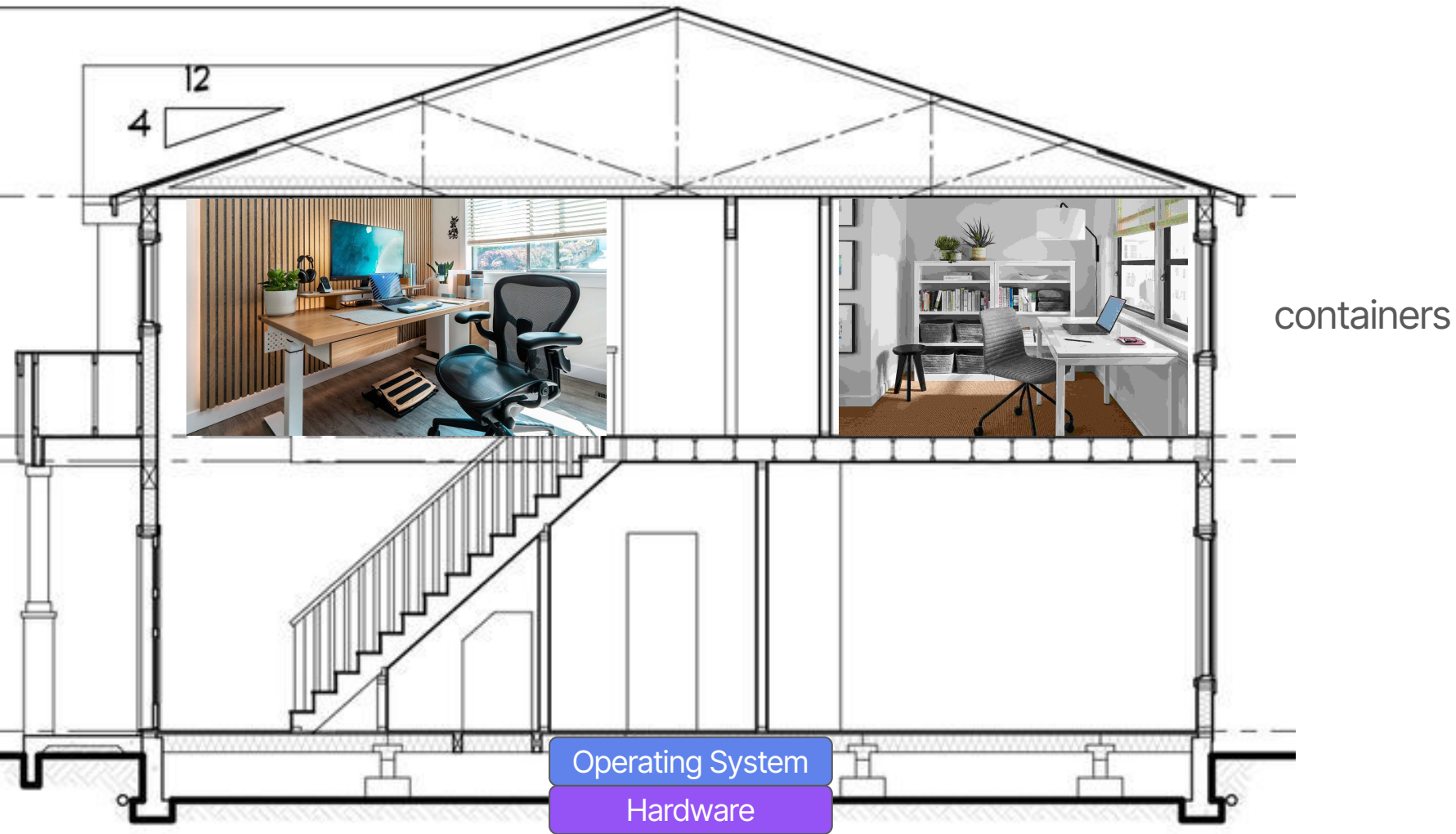
Containers

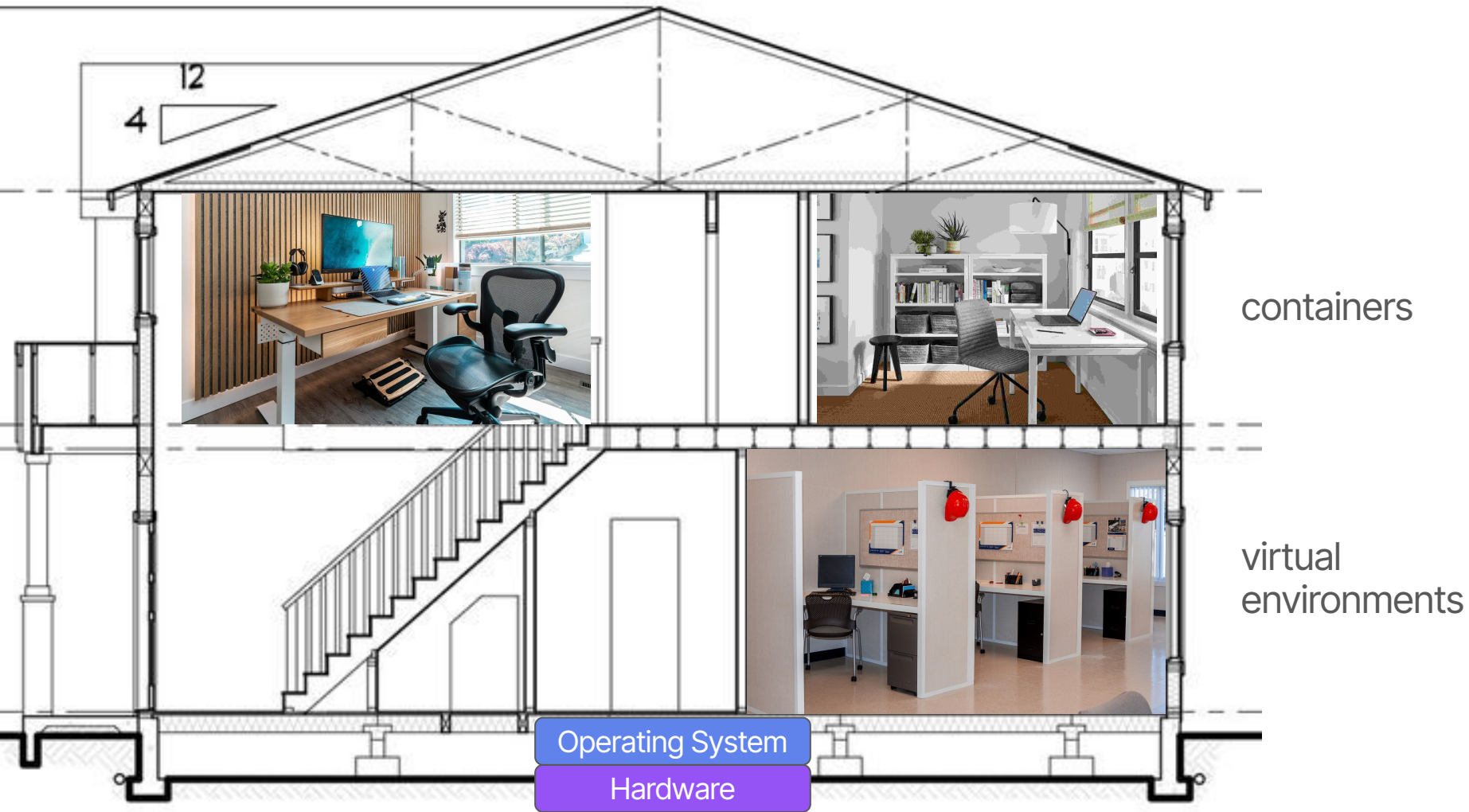




Operating System

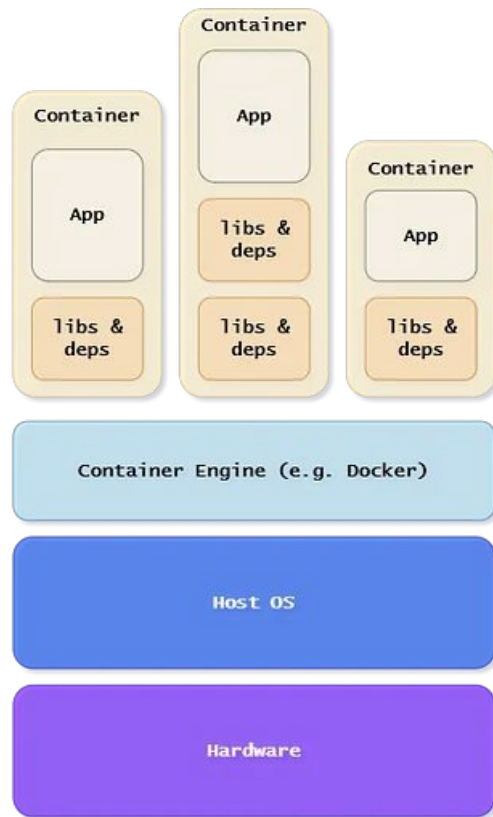
Hardware







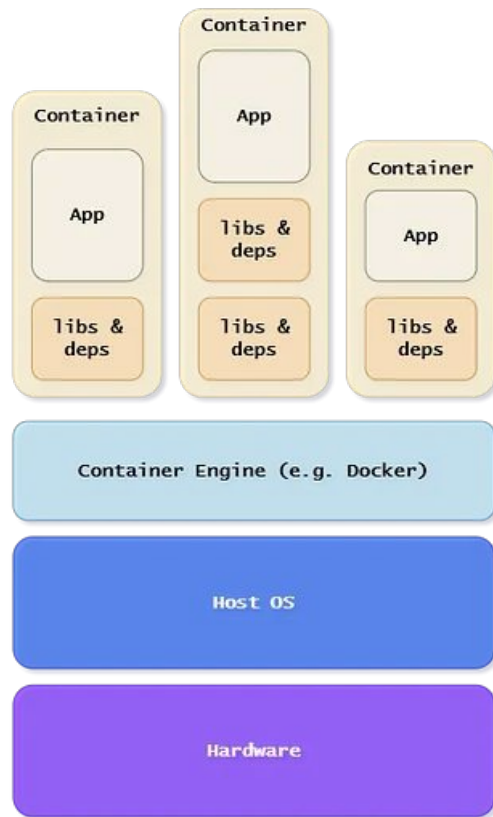
Bare Metal



Containers



Bare Metal

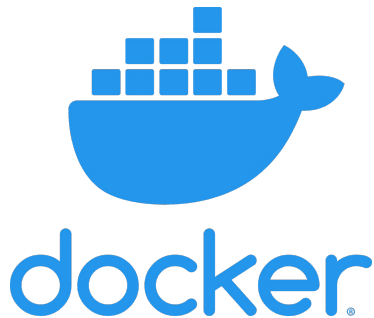


Containers



Virtual Machines

What is Docker?



Docker Engine

- a command line program
- gets and builds Docker images and runs Docker containers



Docker Hub

- a website / web service
- a central repository to store and share Docker container images (commercial)
- other container image registries exist

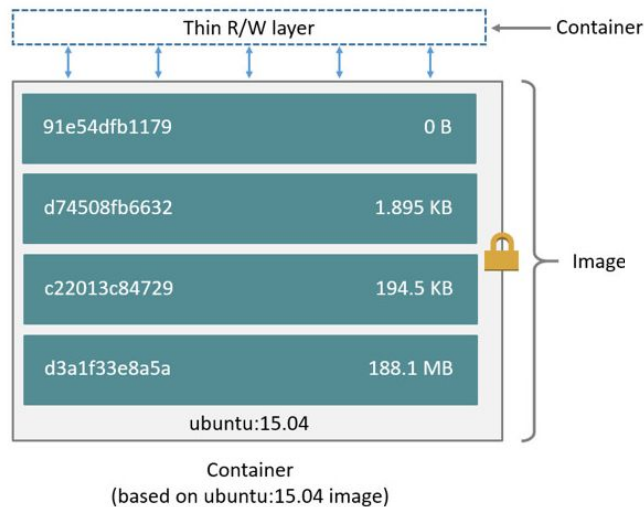
Docker **image** and **container**: what's the difference?

Docker image

- a **read-only** snapshot of an environment
- changing an image adds more layers
- can be stored on Docker Hub or as a file
- images can share identical layers
- can make your own with a **Dockerfile**

Docker container

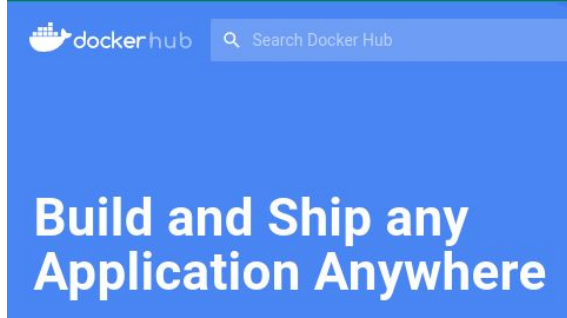
- a **live instance** of a Docker image
- has a thin writable layer that dies with it
- **one image** can spawn **many containers**



How can I get my own Docker container going?

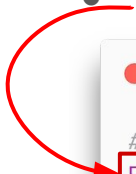
Use an **existing image**

- Docker Hub: a repository of docker images
<https://hub.docker.com/>
- You can pull an image with `docker pull`



Build your **own image**

- a **Dockerfile** lets you describe the exact image you want to create
- Start from one image and add to it

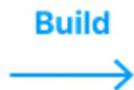


```
# syntax=docker/dockerfile:1
FROM python:3.10-slim-buster
WORKDIR /app
COPY . /app/src
RUN pip install -r /app/src/requirements.txt
RUN pip install --no-cache-dir --no-deps /app/src[all]

ENTRYPOINT [ "bagel" ]
CMD [ "--help" ]
```



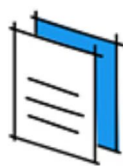
Docker
File



Docker
Image



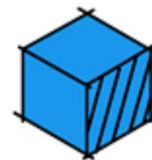
Docker
Container



Docker
File



Docker
Image



Docker
Container



recipe

combine



dough/mix

make
from



pancake
(with toppings)

Exercise 1: Run a container from Docker Hub

- Find an image we like: https://hub.docker.com/_/hello-world
 - Already did this? Try <https://hub.docker.com/r/grycap/cowsay/>
- Pull the image (a specific tag)
- Run the image

Exercise 1: Run a container from Docker Hub

- Find an image we like: https://hub.docker.com/_/hello-world
 - Already did this? Try <https://hub.docker.com/r/grycap/cowsay/>
- Pull the image (a specific tag)
- Run the image

Copy and paste to pull this image

```
docker pull hello-world
```



[View Available Tags](#)

```
surchs@deeptthought:~/Documents/docker_place
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
```



`docker --help` lists
main docker commands!

Exercise 1: Run a container from Docker Hub

Summary

- Docker Hub has pre-made images
- Image tags are important
- We can
 - retrieve images with `docker pull`
 - view images we have pulled with `docker images`
 - (pull and immediately) run a container from an image with `docker run`

Let's find a more useful image

Exercise 2: Work with a container that has conda

I don't have conda installed on my system. But there is a Docker image. Let's try!

- Find a Docker image: <https://hub.docker.com/r/anaconda/miniconda>
- Pick a tag, then pull the image `docker pull anaconda/miniconda:latest`
- Run it

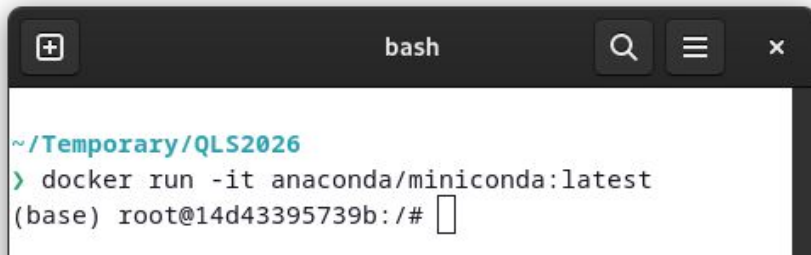
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- Pick a tag, then pull the image `docker pull anaconda/miniconda:latest`
- Run it

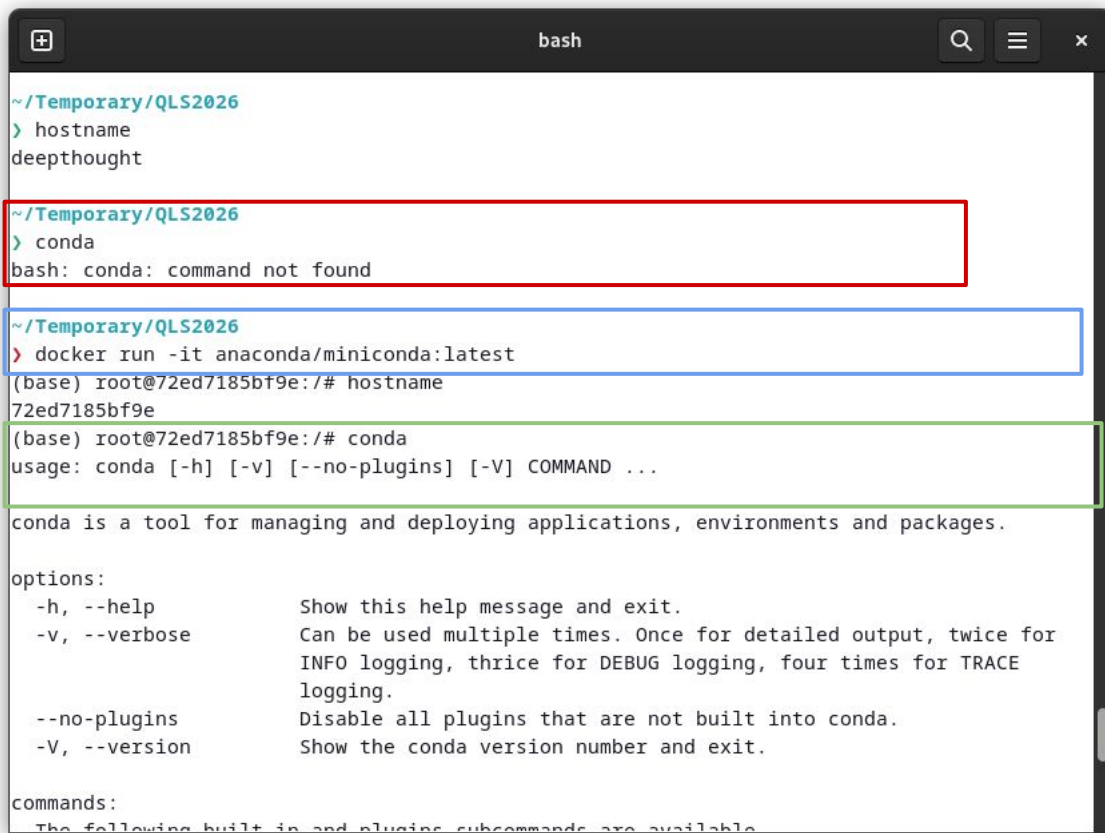
Oh, nothing happened?

- Let's run it interactively to take a look inside

A terminal window with a dark theme. The title bar shows a plus icon, the text 'bash', a search icon, a menu icon, and a close icon. The terminal content shows a prompt '~ /Temporary/QLS2026' followed by the command '> docker run -it anaconda/miniconda:latest'. The next line shows the prompt '(base) root@14d43395739b: /#' with a cursor.

Looking around inside a container

- no conda on my machine
- starting container changes the look of my terminal and the name of my computer
- inside of the container I have access to conda



```
bash

~/Temporary/QLS2026
> hostname
deeptthought

~/Temporary/QLS2026
> conda
bash: conda: command not found

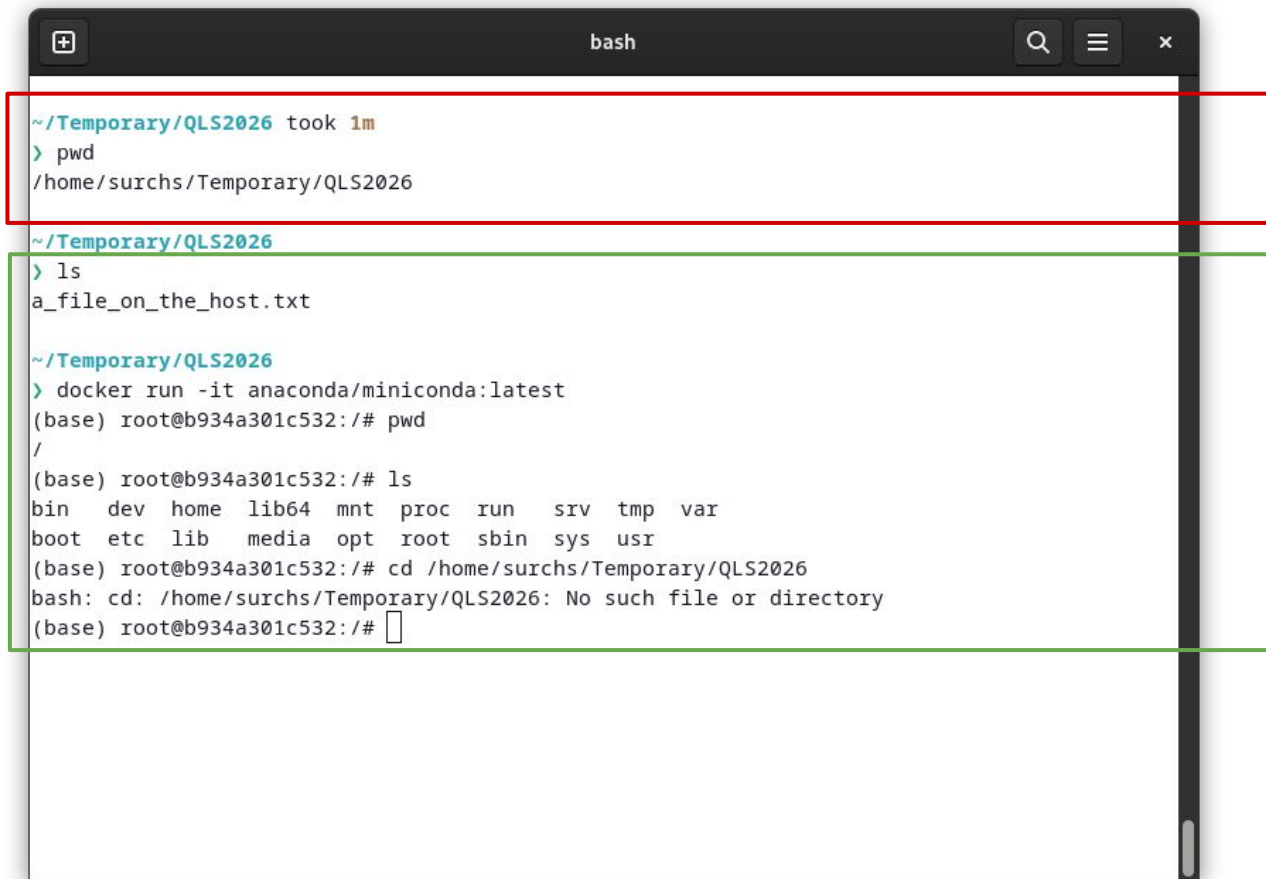
~/Temporary/QLS2026
> docker run -it anaconda/miniconda:latest
(base) root@72ed7185bf9e: /# hostname
72ed7185bf9e
(base) root@72ed7185bf9e: /# conda
usage: conda [-h] [-v] [--no-plugins] [-V] COMMAND ...

conda is a tool for managing and deploying applications, environments and packages.

options:
  -h, --help            Show this help message and exit.
  -v, --verbose          Can be used multiple times. Once for detailed output, twice for
                        INFO logging, thrice for DEBUG logging, four times for TRACE
                        logging.
  --no-plugins          Disable all plugins that are not built into conda.
  -V, --version          Show the conda version number and exit.

commands:
  The following built-in and plugin subcommands are available:
```

By default the container doesn't see files on the host ...



A terminal window titled 'bash' with a search icon, menu icon, and close button. The terminal shows a sequence of commands and outputs. A red box highlights the first two lines, and a green box highlights the subsequent lines.

```
~/Temporary/QLS2026 took 1m
> pwd
/home/surchs/Temporary/QLS2026

~/Temporary/QLS2026
> ls
a_file_on_the_host.txt

~/Temporary/QLS2026
> docker run -it anaconda/miniconda:latest
(base) root@b934a301c532:/# pwd
/
(base) root@b934a301c532:/# ls
bin dev home lib64 mnt proc run srv tmp var
boot etc lib media opt root sbin sys usr
(base) root@b934a301c532:/# cd /home/surchs/Temporary/QLS2026
bash: cd: /home/surchs/Temporary/QLS2026: No such file or directory
(base) root@b934a301c532:/#
```

outside (on host)

inside
(in container)

... and the host can't see files on the container ...

```
bash

~/Temporary/QLS2026
> docker run -it anaconda/miniconda:latest
(base) root@ca6ba83caf1c:/# ls
bin  dev  home  lib64  mnt  proc  run  srv  tmp  var
boot  etc  lib   media  opt  root  sbin  sys  usr
(base) root@ca6ba83caf1c:/# touch a_container_file.txt
(base) root@ca6ba83caf1c:/# ls
a_container_file.txt  boot  etc  lib   media  opt  root  sbin  sys  usr
bin                  dev   home  lib64  mnt    proc  run  srv  tmp  var
(base) root@ca6ba83caf1c:/#
exit

~/Temporary/QLS2026 took 32s
> ls /a_container_file.txt
ls: cannot access '/a_container_file.txt': No such file or directory

~/Temporary/QLS2026
> 
```

inside
(in container)

outside (on host)

... and files written to a specific container are tied to it!

```
bash
~/Temporary/QLS2026
> docker run -it anaconda/miniconda:latest
(base) root@6db1c4b85873:/# ls
bin  dev  home  lib64  mnt  proc  run  srv  tmp  var
boot etc  lib   media  opt  root  sbin  sys  usr
(base) root@6db1c4b85873:/# touch file1.txt
(base) root@6db1c4b85873:/# ls
bin  dev  file1.txt  lib   media  opt  root  sbin  sys  usr
boot etc  home      lib64  mnt   proc  run  srv  tmp  var
(base) root@6db1c4b85873:/#
exit

~/Temporary/QLS2026 took 18s
> docker run -it anaconda/miniconda:latest
(base) root@2240e7358317:/# ls file1.txt
ls: cannot access 'file1.txt': No such file or directory
(base) root@2240e7358317:/# ls
bin  dev  home  lib64  mnt  proc  run  srv  tmp  var
boot etc  lib   media  opt  root  sbin  sys  usr
(base) root@2240e7358317:/#
exit

~/Temporary/QLS2026 took 22s
> □
```

first container instance

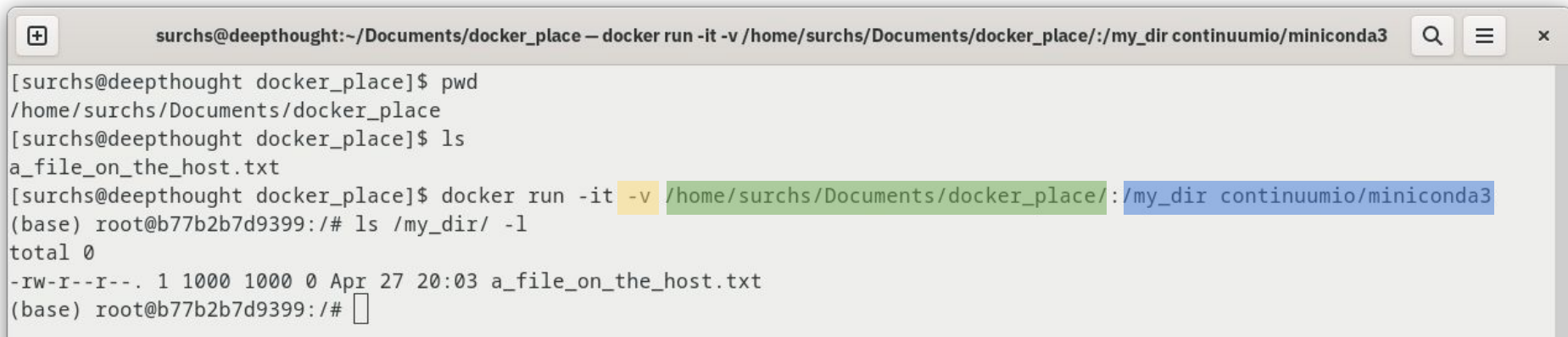
second container instance

Don't write anything (worth keeping) to a container

 `docker ps -a`
lists all containers

How do we share files between host and container?

- Bind mount a path on the host to a path in the container



```
surchs@deeptthought:~/Documents/docker_place — docker run -it -v /home/surchs/Documents/docker_place/:/my_dir continuumio/miniconda3
[surchs@deeptthought docker_place]$ pwd
/home/surchs/Documents/docker_place
[surchs@deeptthought docker_place]$ ls
a_file_on_the_host.txt
[surchs@deeptthought docker_place]$ docker run -it -v /home/surchs/Documents/docker_place/:/my_dir continuumio/miniconda3
(base) root@b77b2b7d9399:/# ls /my_dir/ -l
total 0
-rw-r--r--. 1 1000 1000 0 Apr 27 20:03 a_file_on_the_host.txt
(base) root@b77b2b7d9399:/#
```

- The target path will be created in the container, regardless of if it already exists
- Host path must start with / or . / ! Or use `--mount` instead
- Container path must be an absolute path (start with /)

How do we share files between host and container?

- Bind mount a path on the host to a path in the container

```
surchs@deeptthought:~/Documents/docker_place — docker run -it -v /home/surchs/Documents/docker_place/:/my_dir continuumio/miniconda3
[surchs@deeptthought docker_place]$ pwd
/home/surchs/Documents/docker_place
[surchs@deeptthought docker_place]$ ls
a_file_on_the_host.txt
[surchs@deeptthought docker_place]$ docker run -it -v /home/surchs/Documents/docker_place/:/my_dir continuumio/miniconda3
(base) root@b77b2b7d9399:/# ls /my_dir/ -l
total 0
-rw-r--r--. 1 1000 1000 0 Apr 27 20:03 a_file_on_the_host.txt
(base) root@b77b2b7d9399:/#
```

- The target path will be created in the container, regardless of if it already exists
- Host path must start with / or . / ! Or use `--mount` instead
- Container path must be an absolute path (start with /)

🤔 which user owns files created by the container?

Exercise 2: Work with a container that has conda

Summary

- We can connect to an interactive shell in a container with `docker run -it`
- By default, the container cannot see or write to the filesystem of the host
- We can “mount” a path on the host into the container with

```
docker run -v /host/path:/container/path OR
```

```
docker run --mount type=bind,source=/host/path,target=/container/path
```

- This allows changes to persist beyond the container
- It's a bad idea to write directly into the container's filesystem

Exercise 3: Run a Dockerized tool on local data

We've explored how a Docker container behaves on the inside. Let's follow the more "typical" steps to run a tool available as a Docker image.



inside



outside

Exercise 3: Run a Dockerized tool on local data

Let's try running the [BIDS validator](#) on a test dataset ds006, to see if the dataset complies with the Brain Imaging Data Structure.

1. **Pull** the [bids/validator](#) Docker image tagged latest, and **list** the images on your system to confirm that bids/validator is there

```
docker pull bids/validator:latest  
docker images
```

Exercise 3: Run a Dockerized tool on local data

Recall: `docker run [docker-options] <imagename> <command>`

Print the help text for the bids/validator:

```
docker run bids/validator --help
```

Output:

```
Usage: bids-validator <dataset_directory> [options]
```

...

Exercise 3: Run a Dockerized tool on local data

Recall: `docker run [docker-options] <imagename> <command>`

Print the help text for the bids/validator:

```
docker run bids/validator --help
```

Output:

```
Usage: bids-validator <dataset_directory> [options]
```

...

...How do I run other validator commands?

```
docker run [options] bids/validator <dataset_directory> [options]
```

Exercise 3: Run a Dockerized tool on local data

```
docker run [options] bids/validator <dataset_directory> [options]
```

Our test dataset is located on our **host machine** under:

~/QLS-course-materials/Lectures/2026/data/ds006

2. Run a bids/validator container again, but provide this dataset directory to validate.

Exercise 3: Run a Dockerized tool on local data

```
docker run [options] bids/validator <dataset_directory> [options]
```

Our test dataset is located on our **host machine** under:

```
~/QLS-course-materials/Lectures/2026/data/ds006
```

2. Run a bids/validator container again, but provide this dataset directory to validate.

HINTS:

- Mount the dataset directory into the container for it to find the dataset (remember: the path must start with / or ./!)
`-v /path/on/host:/path/in/container`
- Where is the bids/validator app actually running? What files can it see?

Exercise 3: Run a Dockerized tool on local data

```
docker run [options] bids/validator <dataset_directory> [options]
```

Our test dataset is located on our **host machine** under:

```
~/QLS-course-materials/Lectures/2026/data/ds006
```

2. Run a bids/validator container again, but provide this dataset directory to validate.

HINTS:

- Mount the dataset directory into the container for it to find the dataset (remember: the path must start with / or ./!)
`-v /path/on/host:/path/in/container`
- Where is the bids/validator app actually running? What files can it see?

```
cd ~/QLS-course-materials/Lectures/2026/data
```

```
docker run -v ./ds006:/data bids/validator /data
```

Exercise 3: Run a Dockerized tool on local data

The BIDS validator returns errors [ERR] because the test dataset contains empty data files.

3. To make the validator happy, run it again on the dataset, but this time add **command options** to:

- Ignore warnings
- Ignore NIfTI header content

Exercise 3: Run a Dockerized tool on local data

The BIDS validator returns errors [ERR] because the test dataset contains empty data files.

3. To make the validator happy, run it again on the dataset, but this time add **command options** to:

- Ignore warnings
- Ignore NIfTI header content

HINTS:

- Print the help text again to see all command options for bids/validator
`docker run --rm bids/validator --help`

Exercise 3: Run a Dockerized tool on local data

The BIDS validator returns errors [ERR] because the test dataset contains empty data files.

3. To make the validator happy, run it again on the dataset, but this time add **command options** to:

- Ignore warnings
- Ignore NIfTI header content

HINTS:

- Print the help text again to see all command options for bids/validator

```
docker run --rm bids/validator --help
```

```
docker run -v ./ds006:/data bids/validator /data --ignoreNiftiHeaders  
--ignoreWarnings
```

Apptainer (Singularity)



APPTAINER

Apptainer is the container solution for HPCs (e.g. Compute Canada)

On shared systems (like a supercomputer), you shouldn't/can't use Docker

- Docker isn't as isolated as a VM
- By default you run docker with root privileges and are `root` inside a container
- A malicious actor can escalate privileges and "break out" of a container

Instead, **Apptainer** (formerly Singularity) can be used to run containers:

Singularity^[1] is open source software created by Berkeley Lab:

- as a **secure way** to use Linux containers on Linux multi-user clusters,
- as a way to enable users to have **full control of their environment**, and,
- as a way to **package scientific software** and deploy such to *different* clusters having the *same* architecture.

i.e., it provides **operating-system-level virtualization** commonly called *containers*.



Singularity and Apptainer commands are nearly identical

Build images with Docker, run them with Apptainer

1. Pull an image from a Docker registry and create a local **SingularityImageFile** (.sif)

```
$ apptainer pull docker://ghcr.io/apptainer/lolcow
```

2. Run the Singularity File using `apptainer run`

can also be Docker Hub

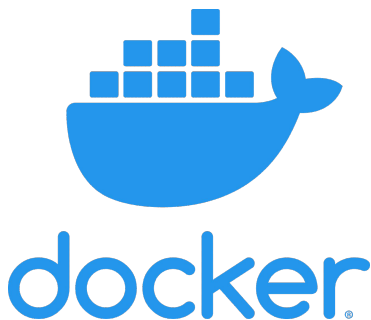
```
$ apptainer run lolcow_latest.sif
```

```
< Mon Aug 16 13:01:55 CDT 2021 >
```

```
-----  
  \  ^__^  
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      (__)\\       )\\/\  
          ||----w |  
          ||     ||
```

Containerization summary

- Docker **images** are **read-only snapshots**, you can find them on Docker Hub
- Images have tags (or versions), that you should specify when pulling
- A Docker **container** is an isolated, **live instance** of an image
- Docker **containers** have their own filesystem, but this is **not persistent**
- With **bind mounts** we can **expose directories** on the host to the container
- We can **build** our own images on top of existing ones using a **Dockerfile**
- Use **Apptainer** to run Docker images on a compute cluster



Docker engine

```
1 # our base image
2 FROM alpine:3.5
3
4 # Install python and pip
5 RUN apk add --update py2-pip
6
7 # upgrade pip
8 RUN pip install --upgrade pip
9
10 # install Python modules needed by the Python app
11 COPY requirements.txt /usr/src/app/
```

Dockerfile



Docker image registry

There are tools to help make Dockerfiles

Welcome to Neurodocker!

Neurodocker is a command-line program that generates custom Dockerfiles and Singularity recipes for neuroimaging and minimizes existing containers. Its purpose is to make it easier for scientists (and others) to easily create reproducible computational environments.

This website is in progress. Is there is something you would like to see here, please [submit a new GitHub issue](#).

```
neurodocker generate docker \  
  --pkg-manager apt \  
  --base-image neurodebian:bullseye \  
  --ants version=2.4.3 \  
  --miniconda version=latest conda_install="nipype notebook" \  
  --user nonroot
```

Additional resources

- Docker docs <https://docs.docker.com/get-started/>
- Hands-on docker tutorials <https://training.play-with-docker.com/>,
<https://www.docker.com/101-tutorial/>
- Neurohackweek container course
<https://neurohackweek.github.io/docker-for-scientists/>
- The Turing Way on reproducible research environments
<https://the-turing-way.netlify.app/reproducible-research/renv.html>
- Russ Poldrack's book "Better Code, Better Science"
<https://bettercodebetterscience.github.io/book/project-organization/#containers>
- Podman (a daemonless docker alternative) <https://docs.podman.io/en/latest/>